

Melkamu Hunegnaw Asmare, PhD

Health AI · Affordable health technology · Digital health

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[KU Leuven profile](#) · [Google Scholar](#) · [ORCID: 0000-0003-0990-3348](#)

Research profile

Researcher and educator working at the intersection of health AI, biomedical signal processing, affordable health technology and equitable digital health. Experience across research, teaching and international partnerships in Africa, Asia and Europe. Research interests include machine learning for screening and diagnosis, responsible AI, health data governance and implementation in resource-constrained settings.

Academic experience

Postdoctoral Researcher & Research Coordinator · 2023–present

Leuven Centre for Affordable Health Technology / eMedia Research Lab, KU Leuven. Research and proposal development, project co-leadership, international partnerships, centre coordination and teaching.

Assistant Professor · 2022–2023

Addis Ababa Institute of Technology, Addis Ababa University. Teaching in health informatics, research ethics, biomedical instrumentation and programming; MSc and BSc supervision.

Online Program Director, part-time · 2022–2023

American College of Technology, Addis Ababa. Online graduate programme coordination and development of an MSc curriculum in Software Engineering.

Doctoral Researcher · 2017–2022

STADIUS, KU Leuven. Biosignals and machine learning for rheumatic heart disease detection, teaching assistance and MSc supervision.

Lecturer · 2014–2017

Addis Ababa Institute of Technology, Addis Ababa University. Biomedical instrumentation, image processing, biomedical signals and research methods.

Education

PhD, Engineering Technology · KU Leuven, Belgium · 2022

Specialization: Health Innovation (Biomedical Engineering)

[Center for Dynamical Systems, Signal Processing, and Data Analytics \(STADIUS\)](#)

Thesis: *Rheumatic Heart Disease Detection*. Supervisor: Prof. Bart Vanrumste.

MSc, Electrical and Electronic Engineering · Universiti Teknologi PETRONAS, Malaysia · 2009. Specialisation: Digital Image Processing.

BSc, Electrical and Computer Engineering · Addis Ababa University, Ethiopia · 2005. Specialisation: Computer Engineering.

Additional qualification: Higher Diploma in Learning-Teaching in Higher Education, Addis Ababa University.

Research & teaching

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Selected project portfolio

Proposal developer and co-PI / project co-promoter. Descriptions below summarise project aims.

JustHealth · Horizon Europe MSCA Doctoral Network

Ethics and governance for just and trustworthy healthcare AI. Doctoral training and clinical applications, including rheumatic heart disease screening and staging.

DiaLink Palestine · VLIR-UOS TEAM · 2026–2031

Partnership with Palestine Polytechnic University on mobile support for Type 2 diabetes care, explainable clinical decision support and a Digital Health Living Lab.

TRUST · VLIR-UOS TEAM · 2026–2031

Digital mental health support for war-affected youth, in partnership with Mekelle and Aksum universities. Participatory design, safeguarding and data governance.

EQUIP · VLIR-UOS TEAM · 2026–2031

Accessible micro-certifications and inclusive training for frontline health and agriculture workers, with Arba Minch University and Arba Minch College of Teacher Education.

AISLE · VLIR-UOS Short Initiative · 2025–2027

AI-assisted and tactile learning tools for visually impaired students in Tigray, with Mekelle University.

Global Affordable Health Technology Week at Group T

KU Leuven Global Minds Open Faculty initiative connecting international university and industry partners around accessible healthcare, education and research.

Teaching & supervision

KU Leuven: Sensors and Circuits for Healthcare Applications; teaching / co-teaching Health Engineering Experience, a project-based Master's course.

Addis Ababa University: Health Informatics, Research Methods, Research Ethics, Biomedical Signals, Medical Image Processing, Biomedical Instrumentation, Advanced Biomedical Instrumentation and Object-Oriented Programming.

Supervision: Five PhD co-promoter roles: three at KU Leuven and two at Addis Ababa University. Additional KU Leuven roles: unofficial co-promoter for Solomon Gebru Abay (2022–2026) and assessor for Amsalu Tomas Chuma (2022–2026; PhD defended in 2026). Master's and Bachelor's research supervision and advising at both institutions.

Authored teaching manuscripts · unpublished

- *Research Ethics in Biomedical Engineering: Principles, Practice, and Equity for Responsible Health Technology* (2026).
- *Sensors and Circuits for Healthcare Applications: Sensors, Analog Front Ends, Signal Conditioning, and Medical Measurement Systems* (2026).

Publications & service

Melkamu Hunegnaw Asmare · Academic CV

Selected publications

1. Abay, S. G., **Asmare, M. H.**, Jacob, J., & Geurts, L. (2026). Clinical validation of an ultra-low-cost smartphone-based offline AI platform for glaucoma screening in low-resource settings. *BMC Medical Informatics and Decision Making*. Advance online publication.
[doi:10.1186/s12911-026-03829-y](https://doi.org/10.1186/s12911-026-03829-y)
2. Abay, S. G., **Asmare, M. H.**, Jacob, J., & Geurts, L. (2026). Deep Learning-Based Glaucoma Screening Using Smartphone-Captured Low-Quality Fundus Videos. *IEEE Access*, 14, 90770–90786.
[doi:10.1109/ACCESS.2026.3702862](https://doi.org/10.1109/ACCESS.2026.3702862)
3. Chuma, A. T., Kassie, D. M., Voigt, J.-U., **Asmare, M. H.**, et al. (2026). Prevalence of Early Rheumatic Heart Disease Among Asymptomatic Students in Underserved Communities in Ethiopia: Cross-Sectional Observational Study. *JMIR Public Health and Surveillance*, 12, e87039.
[doi:10.2196/87039](https://doi.org/10.2196/87039)
4. Chuma, A. T., **Asmare, M. H.**, Varon, C., Kassie, D. M., Wang, C., & Vanrumste, B. (2026). Machine learning for classification of advanced rheumatic heart disease using electrocardiogram in cardiology ward. *BMC Cardiovascular Disorders*, 26, 6. First online publication: 2025.
[doi:10.1186/s12872-025-05315-1](https://doi.org/10.1186/s12872-025-05315-1)
5. **Asmare, M. H.**, Chuma, A. T., Varon, C., Bayou, F. W., Janssens, L., & Vanrumste, B. (2023). Characterization of rheumatic heart diseases from electrocardiogram recordings. *Physiological Measurement*, 44(2), 025002.
[doi:10.1088/1361-6579/aca6cb](https://doi.org/10.1088/1361-6579/aca6cb)
6. **Asmare, M. H.**, Filtjens, B., Woldehanna, F., Janssens, L., & Vanrumste, B. (2021). Rheumatic Heart Disease Screening Based on Phonocardiogram. *Sensors*, 21(19), 6558.
[doi:10.3390/s21196558](https://doi.org/10.3390/s21196558)

Academic service & recognition

- Member, Examination Committee, Faculty of Engineering Technology, Campus Group T, KU Leuven; Digital Society Institute; [Task & Finish Group on Open Science \(2025–2026\)](#).
- Guest Professor, BINUS University, Indonesia.
- Future Digital Leader, Digital Futures, KTH, Sweden (2025).
- Keynote speaker, International Conference on Robotics, Intelligent Systems and Artificial Intelligence, Ethiopia (2024).
- University Medal, Addis Ababa University Research Week (2019).